Concept 1 | Fundamentals of Random Variables

English Student Edition • Focused formulas and guided practice

Formula opener

Expectation

$$E(X) = \sum xp(x)$$

Variance

$$\mathrm{Var}(X) = E(X^2) - [E(X)]^2$$

Probability

$$\sum p(x) = 1$$

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Worked Solutions **Fundamentals of Random Variables**

Q001 Written

For the sum of two dice, construct the probability distribution and find P5 $<= X <= 8$.

Solution

1. List the 36 equally likely pairs, count each sum, and divide each frequency by 36.

Final answer

Values 2 to 12 have frequencies 1,2,3,4,5,6,5,4,3,2,1 over 36; P5 $<= X <=$ 8=20/36=5/9.

Q002 Written

Find the probability distributions for the absolute difference of two dice and the number of white balls drawn from the stated bag; evaluate the requested interval probabilities.

Solution

1. Check that all table probabilities add to 1 before evaluating an interval event.

Final answer

Use the sample space or combinations to list every possible X and sum the relevant probabilities.

Worked Solutions **Fundamentals of Random Variables**

Q003 Written

Construct both source probability tables (four coins; three balls from 4 red and 2 blue) and evaluate the given ranges.

Solution

1. A probability distribution is a table of every possible value of X and its probability.

Final answer

Use binomial or hypergeometric counts, then add probabilities for the requested values.

Q004 Written

Find EX for tails in four coins and for red balls in a draw of three from a bag with 4 red and 5 white.

Solution

1. Build the table, multiply each value by its probability, and add the products.

Final answer

For four fair coins EX=2; for the hypergeometric draw EX=4/3.

Worked Solutions **Fundamentals of Random Variables**

Q005 Written

Find expected values for tails in three coins, the absolute difference of two dice, and white balls in the stated bag.

Solution

1. Keep the random variable definition unchanged while forming the table.

Final answer

Use EX=sum xP$X = x$ after constructing each distribution.

Q006 Written

Find EX for all six source random-variable experiments (cards, dice, balls, tickets, coins and repeated die rolls).

Solution

1. The expected number of successes in repeated identical trials can be checked with n times p.

Final answer

Compute each mean from its probability distribution; symmetry gives quick checks for fair cards and dice.

Worked Solutions **Fundamentals of Random Variables**

Q007 **Written**

For tails in four fair coins, find E(3X+2), E(-4X+1), and EX^2.

Solution

1. Constants shift the mean; a multiplier scales the mean by the same factor.

Final answer

Use $E(aX + b) = aEX$+b; calculate EX^2 directly from the table.

Q008 **Written**

For white balls drawn from 4 red and 3 white, find the four requested transformed expected values.

Solution

1. Do not replace EX^2 by EX^2; they are different quantities.

Final answer

Apply linearity for aX+b and use the weighted square sum for X^2.

Worked Solutions **Fundamentals of Random Variables**

Q009 Written

Solve both source Exercise groups of transformed expectations for coins and for balls.

Solution

1. Write the transformation before substituting the known mean.

Final answer

For every expression use $E(aX + b) = aEX$+b; evaluate powers from the distribution table.

Q010 Written

For black balls drawn from the stated box, find EX and VX.

Solution

1. The distribution is P$X = 0$=2/7, P$X = 1$=4/7, P$X = 2$=1/7. Hence EX=6/7 and VX=EX^2-E$X^2 = 20/49$.

Final answer

$EX = 6/7; VX = 20/49.$

Worked Solutions **Fundamentals of Random Variables**

Q011 Written

Find the mean and variance for the two source Try random variables (a numbered card and winning tickets).

Solution

1. Variance measures spread around the mean, not the mean itself.

Final answer

Construct each probability table, then use VX=EX^2-EX^2.

Q012 Written

Find EX and VX for all five source Exercise random variables.

Solution

1. For sampling without replacement, use combinations to obtain the exact probabilities.

Final answer

Use the distribution table for each experiment and the variance identity.

Worked Solutions **Fundamentals of Random Variables**

Q013 Written

Find the standard deviation of tails when two biased coins have P(head)=4/7 and P(tail)=3/7.

Solution

1. Find VX first, then take the positive square root.

Final answer

The table gives VX=24/49, so sigma$X$=2 sqrt6/7.

Q014 Written

Find the standard deviation from each source probability table and for white balls drawn from 4 red and 3 white.

Solution

1. Standard deviation has the same units as X and is always non-negative.

Final answer

Use sigma$X$=sqrt$V(X)$ after evaluating the variance from the table.

Worked Solutions **Fundamentals of Random Variables**

Q015 Written

Find the standard deviation for both source probability-table exercises.

Solution

1. Keep the positive square root only.

Final answer

Compute EX, EX^2, then sigma$X$=sqrt$E(X^2) - E(X)^2$.

Q016 Written

For the number of winning tickets in the source draw, find VY and sigma$Y$ for Y=2X+1 and Y=-3X-2.

Solution

1. The constant b changes neither variance nor standard deviation.

Final answer

Use $V(aX + b) = a^2VX$ and sigma(aX+b)=|a|sigma$X$.

Worked Solutions **Fundamentals of Random Variables**

Q017 Written

For red balls drawn from the stated bag, find variance and standard deviation for the three linear transformations of X.

Solution

1. Write both transformation rules before substituting numbers.

Final answer

Apply the squared multiplier to VX and the absolute multiplier to sigma$X$.

Q018 Written

For a single die, find VY and sigma$Y$ for Y=X+2, Y=3X+1 and Y=-2X+4.

Solution

1. A translation changes the mean but leaves spread unchanged.

Final answer

Use the variance and standard-deviation transformation formulas for each Y.

Worked Solutions **Fundamentals of Random Variables**

Q019 **Written**

Solve the two source transformation problems: a net prize after an entry fee, and finding a,b from E(aX+b)=7/3 and sigma(aX+b)=3.

Solution

1. The mean equation determines b after the standard-deviation equation fixes a.

Final answer

Use $E(aX+b)=aEX$+b and sigma(aX+b)=|a|sigma$X$, with a>0.

Q020 **Written**

Solve the two source Try problems involving net winnings and determining a,b for a die transformation.

Solution

1. Entry fees are negative constants; they shift the mean only.

Final answer

Model the payout as $Y=aX+b$, then apply the mean and standard-deviation rules.

Worked Solutions **Fundamentals of Random Variables**

Q021 Written

Solve all four source transformation exercises, including the number-line coin walk, games, and the calibrated die model.

Solution

1. Translate the words into aX+b before calculating.

Final answer

Express every quantity as $Y = aX + b$ and use the appropriate mean, variance or standard-deviation rule.

Q022 MCQ

A probability distribution must have probabilities that sum to:

Solution

1. Check the table total.

Correct option: B

Final answer

1

Worked Solutions **Fundamentals of Random Variables**

Q023 MCQ

For two fair dice, P$X = 2$ is:

Solution

1. Only the pair (1,1) gives sum 2.

Correct option: B

Final answer

1/36

Q024 MCQ

The number of heads in three coins can be:

Solution

1. Count heads in the sample space.

Correct option: B

Final answer

0, 1, 2, 3

Worked Solutions **Fundamentals of Random Variables**

Q025 MCQ

The expected value is calculated as:

Solution

1. Multiply each value by its probability.

Correct option: B

Final answer

sum x$PX = x$

Q026 MCQ

The mean number of tails in four fair coins is:

Solution

1. Use symmetry or the table.

Correct option: B

Final answer

2

Worked Solutions **Fundamentals of Random Variables**

Q027 MCQ

For a fair die, EX is:

Solution

1. The values 1 to 6 average to 3.5.

Correct option: B

Final answer

3.5

Q028 MCQ

E(aX+b) equals:

Solution

1. Use linearity of expectation.

Correct option: B

Final answer

$aEX + b$

Worked Solutions **Fundamentals of Random Variables**

Q029 MCQ

In general, $\mathrm{E}X^2$ is:

Solution

1. Square values before weighting.

Correct option: C

Final answer

weighted square sum

Q030 MCQ

If $\mathrm{E}X$=2, then E(3X+2) is:

Solution

1. Calculate 32+2.

Correct option: C

Final answer

8

Worked Solutions **Fundamentals of Random Variables**

Q031 MCQ

The variance identity is:

Solution

1. Subtract the squared mean.

Correct option: B

Final answer

$EX^2 - EX^2$

Q032 MCQ

Variance can never be:

Solution

1. It is an expected square.

Correct option: C

Final answer

negative

Worked Solutions **Fundamentals of Random Variables**

Q033 MCQ

The first step in a variance problem is to:

Solution

1. The table supplies $\mathrm{E}X$ and $\mathrm{E}X^2$.

Correct option: A

Final answer

draw the distribution

Q034 MCQ

Standard deviation is:

Solution

1. Take the positive square root.

Correct option: B

Final answer

sqrt$V(X)$

Worked Solutions **Fundamentals of Random Variables**

Q035 MCQ

Standard deviation has units:

Solution

1. The square root restores the units.

Correct option: B

Final answer

same units as X

Q036 MCQ

V(aX+b) equals:

Solution

1. Constants do not affect spread.

Correct option: B

Final answer

a squared VX

Worked Solutions **Fundamentals of Random Variables**

Q037 MCQ

sigma(aX+b) equals:

Solution

1. Standard deviation is non-negative.

Correct option: B

Final answer

|a| sigma$X$

Q038 MCQ

For Y=X+3, the variance is:

Solution

1. A translation leaves variance unchanged.

Correct option: B

Final answer

VX

Worked Solutions **Fundamentals of Random Variables**

Q039 MCQ

A net prize after a fixed fee is modeled as:

Solution

1. The fee is the constant b.

Correct option: B

Final answer

$aX + b$

Q040 MCQ

If a>0 and sigma(aX+b)=3, then a is:

Solution

1. Solve a sigma$X$=3.

Correct option: B

Final answer

3/sigma$X$

Worked Solutions **Fundamentals of Random Variables**

Q041 MCQ

Changing b in aX+b changes:

Solution

1. b is a translation.

Correct option: C

Final answer

the mean only

Q042 MCQ

If EX=2 and EX^2=7, then VX is:

Solution

1. Calculate $7 - 2^2$.

Correct option: A

Final answer

3

Worked Solutions **Fundamentals of Random Variables**

Quiz WRITTEN 1 **Concept Quiz · Written**

Find EX and VX for the number of heads in three fair coins.

Solution

1. Use the binomial distribution.

Final answer

EX=3/2 and VX=3/4

Quiz WRITTEN 2 **Concept Quiz · Written**

For Y=4X-1 with VX=2, find VY and sigma$Y$.

Solution

1. Use $V(aX + b) = a^2VX$.

Final answer

VY=32 and sigma$Y$=4 sqrt2

Worked Solutions **Fundamentals of Random Variables**

Quiz WRITTEN 3 Concept Quiz · Written

For a fair die, find E(2X+5).

Solution

1. EX=7/2, then use linearity.

Final answer

12

Quiz WRITTEN 4 Concept Quiz · Written

If sigma$X$=2 and EX=3, find sigma(-5X+7).

Solution

1. Multiply standard deviation by |a|.

Final answer

10

Worked Solutions **Fundamentals of Random Variables**

Quiz MCQ 5 **Concept Quiz · MCQ**

If VX=5, V(3X+1) is:

Solution

1. Square the multiplier.

Correct option: D

Final answer

45

Quiz MCQ 6 **Concept Quiz · MCQ**

For a fair die EX is:

Solution

1. Average 1 through 6.

Correct option: C

Final answer

3.5

Worked Solutions **Fundamentals of Random Variables**

Quiz MCQ 7 **Concept Quiz · MCQ**

A probability table total must be:

Solution

1. Add all probabilities.

Correct option: B

Final answer

1

Quiz MCQ 8 **Concept Quiz · MCQ**

If sigma$X$=4, sigma(X-9) is:

Solution

1. Translations do not change spread.

Correct option: B

Final answer

4

Answer key

Use the question number shown on each page.

Q001 Values 2 to 12 have frequencies 1,2,3,4,5,6,5,4,3,2,1 over 36; P5 <= X <= 8=20/36=5/9.

Q002 Use the sample space or combinations to list every possible X and sum the relevant probabilities.

Q003 Use binomial or hypergeometric counts, then add probabilities for the requested values.

Q004 For four fair coins EX=2; for the hypergeometric draw EX=4/3.

Q005 Use EX=sum xP$X = x$ after constructing each distribution.

Q006 Compute each mean from its probability distribution; symmetry gives quick checks for fair cards and dice.

Q007 Use $E(aX + b) = aEX$+b; calculate EX^2 directly from the table.

Q008 Apply linearity for aX+b and use the weighted square sum for X^2.

Q009 For every expression use $E(aX + b) = aEX$+b; evaluate powers from the distribution table.

Q010 $EX = 6/7; VX = 20/49$.

Q011 Construct each probability table, then use VX=EX^2-EX^2.

Q012 Use the distribution table for each experiment and the variance identity.

Q013 The table gives VX=24/49, so sigma$X$=2 sqrt6/7.

Q014 Use sigma$X$=sqrt$V(X)$ after evaluating the variance from the table.

Q015 Compute EX, EX^2, then sigma$X$=sqrt$E(X^2) - E(X)^2$.

Q016 Use $V(aX + b) = a^2VX$ and sigma(aX+b)=|a|sigma$X$.

Q017 Apply the squared multiplier to VX and the absolute multiplier to sigma$X$.

Q018 Use the variance and standard-deviation transformation formulas for each Y.

Q019 Use $E(aX + b) = aEX$+b and sigma(aX+b)=|a|sigma$X$, with a>0.

Q020 Model the payout as $Y = aX + b$, then apply the mean and standard-deviation rules.

Q021 Express every quantity as $Y = aX + b$ and use the appropriate mean, variance or standard-deviation rule.

Q022 B

Q023 B

Q024 B

Q025 B

Q026 B

Q027 B

Q028 B

Q029 C

Q030 C

Q031 B

Q032 C

Q033 A

Q034 B

Q035 B

Q036 B

Q037 B

Q038 B

Q039 B

Q040 B

Q041 C

Q042 A

Quiz WRITTEN 1 EX=3/2 and VX=3/4

Quiz WRITTEN 2 VY=32 and sigma$Y$=4 sqrt2

Quiz WRITTEN 3 12

Quiz WRITTEN 4 10

Quiz MCQ 5 D

Quiz MCQ 6 C

Quiz MCQ 7 B

Quiz MCQ 8 B

Concept 2 | Expected Value and Variance of Multiple Variables

English Student Edition • Focused formulas and guided practice

Formula opener

Linearity

$$E(aX+b)=aE(X)+b$$

Variance

$$\operatorname{Var}(aX+b)=a^2\operatorname{Var}(X)$$

Independent

$$\operatorname{Var}(X+Y)=\operatorname{Var}X+\operatorname{Var}Y$$

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Worked Solutions **Expected Value and Variance of Multiple Variables**

Q001 Written

Given the probability distributions of X and Y in the source tables, find E(X+Y).

Solution

1. Find EX and EY separately, then add them.

Final answer

$EX = 35/3, EY = 9/2, so E(X + Y) = 97/6.$

Q002 Written

Three coins are tossed once; let X, Y and Z be the head indicators. Find E(X+Y+Z).

Solution

1. Each fair head indicator has expected value 1/2.

Final answer

$E(X + Y + Z) = 1/2 + 1/2 + 1/2 = 3/2.$

Worked Solutions **Expected Value and Variance of Multiple Variables**

Q003 **Written**

For the two source tables with X=1,2,3 and Y=4,5,6, find E(X+Y).

Solution

1. The expectation of a sum is the sum of the expectations.

Final answer

$E(X+Y) = EX + EY = 2 + 5 = 7.$

Q004 **Written**

For the two bags in the source exercise, find the expected total number of red balls drawn.

Solution

1. Use E(red drawn)=sample size times the red proportion in each bag.

Final answer

Bag A: 6/7; Bag B: 6/7; total $E = 12/7$.

Worked Solutions Expected Value and Variance of Multiple Variables

Q005 Written

Three independent subsystems each receive two signals with success probability 0.5. Find E(X+Y+Z).

Solution

1. Add the three binomial means.

Final answer

Each mean is $2(0.5) = 1$, so $E(X + Y + Z) = 3$.

Q006 Written

Two 500-point coins and three 100-point coins are tossed. Find the expected total score Z.

Solution

1. Model the score as aX+bY, then apply linearity.

Final answer

$Z = 500X + 100Y, soEZ = 5001 + 1003/2 = 650.$

Worked Solutions **Expected Value and Variance of Multiple Variables**

Q007 Written

Two 500-point coins and two 100-point coins are tossed. Find EZ.

Solution

1. The expected heads in each two-coin group is 1.

Final answer

$EZ = 5001 + 1001 = 600.$

Q008 Written

A die is thrown twice and the score is Z=3X+5Y. Find EZ.

Solution

1. Both die rolls have mean 7/2.

Final answer

$EZ = 37/2 + 57/2 = 28.$

Worked Solutions **Expected Value and Variance of Multiple Variables**

Q009 Written

For three 100-point coins and one 50-point coin, find the expected score from heads.

Solution

1. Use the expected score of each fair coin and add.

Final answer

$EZ = 350 + 25 = 175.$

Q010 Written

In two independent raffles, buy two tickets from a 2/10 winner raffle and two from a 1/6 winner raffle. Find the expected prizes.

Solution

1. Expected winners are sample size times the winning proportion in each raffle.

Final answer

$EZ = 22/10 + 21/6 = 11/15.$

Worked Solutions **Expected Value and Variance of Multiple Variables**

Q011 Written

Three coins are tossed and a die is rolled; X is heads and Y is the die score. Find E(XY).

Solution

1. Check independence before multiplying expectations.

Final answer

X and Y are independent, so $E(XY) = EXEY$=3/27/2=21/4.

Q012 Written

For independent X with probabilities 2/5,2/5,1/5 at 1,2,3 and Y uniform on 2,4,6, find E(XY).

Solution

1. Compute each mean from its table, then multiply.

Final answer

$EX = 8/5, EY = 4, soE(XY) = 32/5.$

Worked Solutions **Expected Value and Variance of Multiple Variables**

Q013 Written

Ahmed tosses three coins and Mohamed tosses two coins. Let X and Y be their head counts. Find E(XY).

Solution

1. The two experiments are independent.

Final answer

$E(XY) = EXEY = 3/21 = 3/2.$

Q014 Written

A card numbered 1 to 4 is replaced and a second card is drawn. Find E(XY).

Solution

1. Replacement makes the two draws independent.

Final answer

$EX = EY = 5/2, soE(XY) = 25/4.$

Worked Solutions **Expected Value and Variance of Multiple Variables**

Q015 Written

For independent X and Y with EX=1 and EY=2/7, find E(XY).

Solution

1. Use the product rule after checking independence.

Final answer

$E(XY) = EXEY = 2/7$.

Q016 Written

A card X is drawn from 1,2,3 and three-coin heads Y is independent. Find V(X+Y) and sigma(X+Y).

Solution

1. For independent variables, add variances, then take the positive square root.

Final answer

VX=2/3, VY=3/4, so $V(X+Y) = 17/12$ and sigma=sqrt17/12.

Worked Solutions **Expected Value and Variance of Multiple Variables**

Q017 Written

Two independent dice are thrown. Find the variance and standard deviation of their sum.

Solution

1. Each fair die has variance 35/12.

Final answer

$V(X+Y) = 35/12 + 35/12 = 35/6$; sigma=sqrt35/6.

Q018 Written

An independent coin head count X and die score Y are combined. Find V(X+Y) and sigma(X+Y).

Solution

1. Use $V(Bernoulli(1/2)) = 1/4$ and $V(die) = 35/12$.

Final answer

$V = 1/4 + 35/12 = 19/6$; sigma=sqrt19/6.

Worked Solutions **Expected Value and Variance of Multiple Variables**

Q019 Written

Ahmed tosses two coins and Mohamed tosses three coins. Find the variance of the total head count.

Solution

1. Variances of independent Bernoulli counts add.

Final answer

$V = 21/4 + 31/4 = 5/4$.

Q020 Written

Independent X is uniform on 1 to 4 and Y on 1 to 6. Find V(X+Y) and sigma(X+Y).

Solution

1. Use the uniform discrete variance for each scale and add.

Final answer

$V = 5/4 + 35/12 = 25/6$; sigma=5/sqrt6.

Worked Solutions **Expected Value and Variance of Multiple Variables**

Q021 Written

Three independent special dice take values 2,4,6 equally. Find E(XYZ) and V(X+Y+Z).

Solution

1. For a product multiply means; for a sum add variances.

Final answer

EX=4, VX=8/3; $E(XYZ) = 64$ and $V(sum) = 8$.

Q022 Written

Three independent dice take values 1,3,5 equally. Find E(XYZ) and V(X+Y+Z).

Solution

1. All three distributions are identical and independent.

Final answer

EX=3, VX=8/3; $E(XYZ) = 27$ and $V(sum) = 8$.

Worked Solutions **Expected Value and Variance of Multiple Variables**

Q023 Written

Three ordinary dice are thrown. Find the expected product and the variance of their sum.

Solution

1. Use independence for the product and variance addition for the sum.

Final answer

$E(product)$ =$7/2^3$ = 343/8; $V(sum)$ = 335/12=35/4.

Q024 Written

Three friends have 5, 4 and 3 independent rolls to score a six. Find E(XYZ) and V(X+Y+Z).

Solution

1. For Binomial(n,1/6), use $E = np$ and $V = np(1 - p)$.

Final answer

Means are 5/6, 2/3, 1/2; $E(product) = 5/18$ and $V(sum) = 5/3$.

Worked Solutions **Expected Value and Variance of Multiple Variables**

Q025 Written

If X, Y and Z are independent head counts from 4, 3 and 2 fair coins, find E(XYZ).

Solution

1. Multiply the three expected head counts.

Final answer

$E(XYZ)$ = 23/21=3.

Q026 MCQ

E(X+Y) equals:

Solution

1. Add the two expectations.

Correct option: B

Final answer

$EX + EY$

Worked Solutions **Expected Value and Variance of Multiple Variables**

Q027 MCQ

For three variables, E(X+Y+Z) equals:

Solution

1. Extend linearity to three terms.

Correct option: B

Final answer

$EX + EY + EZ$

Q028 MCQ

The expected heads in one fair coin toss is:

Solution

1. Use the Bernoulli mean.

Correct option: B

Final answer

1/2

Worked Solutions **Expected Value and Variance of Multiple Variables**

Q029 MCQ

For two signals with p=0.5, EX is:

Solution

1. Use np.

Correct option: B

Final answer

1

Q030 MCQ

If EX=2 and EY=5, E(X+Y) is:

Solution

1. Add the means.

Correct option: B

Final answer

7

Worked Solutions **Expected Value and Variance of Multiple Variables**

Q031 MCQ

E(aX+bY) equals:

Solution

1. Apply linearity.

Correct option: A

Final answer

$aEX + bEY$

Q032 MCQ

The expected score of one fair 100-point coin is:

Solution

1. Half the value is expected.

Correct option: B

Final answer

50

Worked Solutions **Expected Value and Variance of Multiple Variables**

Q033 MCQ

For a fair die, E(3X+5Y) with two rolls is:

Solution

1. Use both means 7/2.

Correct option: C

Final answer

28

Q034 MCQ

A fixed entry fee changes:

Solution

1. It is a constant term.

Correct option: B

Final answer

the mean only

Worked Solutions **Expected Value and Variance of Multiple Variables**

Q035 MCQ

The expected number of winners in 2 of 10 tickets with 2 winners is:

Solution

1. Use sample size times winner proportion.

Correct option: B

Final answer

2/5

Q036 MCQ

E(XY)=EXEY requires:

Solution

1. The product rule needs mutual independence.

Correct option: B

Final answer

independence

Worked Solutions **Expected Value and Variance of Multiple Variables**

Q037 MCQ

For three fair coins, EX is:

Solution

1. Use np.

Correct option: C

Final answer

3/2

Q038 MCQ

For a fair die, EY is:

Solution

1. Average 1 through 6.

Correct option: C

Final answer

3.5

Worked Solutions **Expected Value and Variance of Multiple Variables**

Q039 MCQ

With replacement, two card draws are:

Solution

1. Replacement restores the same distribution and independence.

Correct option: B

Final answer

independent

Q040 MCQ

If EX=2 and EY=3 for independent variables, E(XY) is:

Solution

1. Multiply the means.

Correct option: B

Final answer

6

Worked Solutions **Expected Value and Variance of Multiple Variables**

Q041 MCQ

For independent X,Y, V(X+Y) equals:

Solution

1. Add independent variances.

Correct option: B

Final answer

$VX + VY$

Q042 MCQ

The variance of a fair die is:

Solution

1. Use the standard die variance.

Correct option: B

Final answer

35/12

Worked Solutions **Expected Value and Variance of Multiple Variables**

Q043 MCQ

Sigma(X+Y) is:

Solution

1. Take the positive root.

Correct option: B

Final answer

sqrt$V(X+Y)$

Q044 MCQ

The variance of the sum of two independent fair dice is:

Solution

1. Double 35/12.

Correct option: B

Final answer

35/6

Worked Solutions **Expected Value and Variance of Multiple Variables**

Q045 *MCQ*

For independent sums, a constant shift changes:

Solution

1. Translations do not change spread.

Correct option: C

Final answer

neither spread measure

Q046 *MCQ*

For independent X,Y,Z, E(XYZ) equals:

Solution

1. Multiply the three means.

Correct option: B

Final answer

$EXEYEZ$

Q047 MCQ

For independent X,Y,Z, V(X+Y+Z) equals:

Solution

1. Add the three variances.

Correct option: B

Final answer

$VX + VY + VZ$

Q048 MCQ

For a binomial count with n=5 and p=1/6, EX is:

Solution

1. Use np.

Correct option: B

Final answer

5/6

Worked Solutions **Expected Value and Variance of Multiple Variables**

Q049 MCQ

For three values 2,4,6 equally likely, EX is:

Solution

1. Average the three values.

Correct option: C

Final answer

4

Q050 MCQ

For three independent fair dice, V(sum) is:

Solution

1. Add three copies of 35/12.

Correct option: C

Final answer

35/4

Worked Solutions **Expected Value and Variance of Multiple Variables**

Quiz WRITTEN 1 **Concept Quiz · Written**

If EX=4 and EY=7, find E(3X-2Y).

Solution

1. Use linearity: 34-27.

Final answer

-2

Quiz WRITTEN 2 **Concept Quiz · Written**

For independent variables with EX=2 and EY=5, find E(XY).

Solution

1. Multiply the means.

Final answer

10

Worked Solutions **Expected Value and Variance of Multiple Variables**

Quiz WRITTEN 3 Concept Quiz · Written

Two independent fair dice are thrown. Find V(X+Y) and sigma(X+Y).

Solution

1. Add the two die variances.

Final answer

V = 35/6, sigma=sqrt35/6

Quiz WRITTEN 4 Concept Quiz · Written

Three independent Binomial counts have (n,p)=(2,1/2),(3,1/2),(4,1/2). Find V(X+Y+Z).

Solution

1. Add np(1-p) for the three counts.

Final answer

9/4

Worked Solutions **Expected Value and Variance of Multiple Variables**

Quiz MCQ 5 **Concept Quiz · MCQ**

If EX=3 and EY=4, E(2X+Y) is:

Solution

1. Compute 23+4.

Correct option: B

Final answer

10

Quiz MCQ 6 **Concept Quiz · MCQ**

For independent variables EX=2, EY=5, E(XY) is:

Solution

1. Multiply the means.

Correct option: B

Final answer

10

Worked Solutions **Expected Value and Variance of Multiple Variables**

Quiz MCQ 7 **Concept Quiz · MCQ**

For two independent die rolls, V(X+Y) is:

Solution

1. Add the two variances.

Correct option: B

Final answer

35/6

Quiz MCQ 8 **Concept Quiz · MCQ**

For three independent variables, V(X+Y+Z) is:

Solution

1. Extend variance addition.

Correct option: B

Final answer

the sum of variances

Answer key

Use the question number shown on each page.

Q001 $EX = 35/3, EY = 9/2, soE(X+Y) = 97/6.$

Q002 $E(X+Y+Z) = 1/2 + 1/2 + 1/2 = 3/2.$

Q003 $E(X+Y) = EX + EY = 2 + 5 = 7.$

Q004 Bag A: 6/7; Bag B: 6/7; total $E = 12/7$.

Q005 Each mean is $2(0.5) = 1$, so $E(X+Y+Z) = 3$.

Q006 $Z = 500X + 100Y, soEZ = 5001 + 1003/2 = 650.$

Q007 $EZ = 5001 + 1001 = 600.$

Q008 $EZ = 37/2 + 57/2 = 28.$

Q009 $EZ = 350 + 25 = 175.$

Q010 $EZ = 22/10 + 21/6 = 11/15.$

Q011 X and Y are independent, so $E(XY) = EX$EY=3/27/2=21/4.

Q012 $EX = 8/5, EY = 4, soE(XY) = 32/5.$

Q013 $E(XY) = EXEY = 3/21 = 3/2.$

Q014 $EX = EY = 5/2, soE(XY) = 25/4.$

Q015 $E(XY) = EXEY = 2/7.$

Q016 VX=2/3, VY=3/4, so $V(X+Y) = 17/12$ and sigma=sqrt17/12.

Q017 $V(X+Y) = 35/12 + 35/12 = 35/6$; sigma=sqrt35/6.

Q018 $V = 1/4 + 35/12 = 19/6$; sigma=sqrt19/6.

Q019 $V = 21/4 + 31/4 = 5/4.$

Q020 $V = 5/4 + 35/12 = 25/6$; sigma=5/sqrt6.

Q021 EX=4, VX=8/3; $E(XYZ) = 64$ and $V(sum) = 8$.

Q022 EX=3, VX=8/3; $E(XYZ) = 27$ and $V(sum) = 8$.

Q023 $E(product)$ =7/2^3 $= 343/8$; $V(sum) = 335/12$=35/4.

Q024 Means are 5/6, 2/3, 1/2; $E(product) = 5/18$ and $V(sum) = 5/3$.

Q025 $E(XYZ) = 23/21$=3.

Q026 B

Q027 B

Q028 B

Q029 B

Q030 B

Q031 A

Q032 B

Q033 C

Q034 B

Q035 B

Q036 B

Q037 C

Q038 C

Q039 B

Q040 B

Q041 B

Q042 B

Q043 B

Q044 B

Q045 C

Q046 B

Q047 B

Q048 B

Q049 C

Q050 C

Quiz WRITTEN 1 -2

Quiz WRITTEN 2 10

Quiz WRITTEN 3 $V = 35/6$, sigma=sqrt35/6

Quiz WRITTEN 4 9/4

Quiz MCQ 5 B

Quiz MCQ 6 B

Quiz MCQ 7 B

Quiz MCQ 8 B